📘 **AI/ML Internship – Day 8 Notes & Tasks**

🟢 **Module 3: Python for AI**

📌 **Part 1: Functions**

🔹 **What is a Function?**

A function is a reusable block of code that performs a specific task.

🔹 **Types of Functions**

- Built-in functions → print(), len()
- User-defined functions

🔹 **Function with Parameters**

```
def greet(name):
    print("Hello", name)

greet("AI Student")
```

🔹 **Function with Return Value**

```
def add(a, b):
    return a + b

result = add(5, 3)
print(result)
```

🔹 **Default Parameters**

```
def greet(name="Student"):
    print("Hello", name)

greet()
```

🔹 Keyword Arguments

```
def info(name, age):
    print(name, age)

info(age=20, name="John")
```

📌 Part 2: Recursion

🔹 What is Recursion?

A function calling itself is called recursion.

🔹 Example: Factorial

```
def factorial(n):
    if n == 1:
        return 1
    return n * factorial(n-1)

print(factorial(5))
```

🔹 Key Points

- Must have a **base case**
- Used in complex problems
- Important in algorithms

📌 Part 3: Lambda Functions

🔹 What is Lambda?

A small anonymous function.

```
add = lambda a, b: a + b
print(add(3, 4))
```

🔹 Use with List

```
nums = [1, 2, 3, 4]

squares = list(map(lambda x: x*x, nums))
print(squares)
```

📌 Part 4: Map, Filter (Important for AI)

🔹 map()

```
nums = [1, 2, 3]

result = list(map(lambda x: x*2, nums))
print(result)
```

🔹 filter()

```
nums = [1, 2, 3, 4]

result = list(filter(lambda x: x % 2 == 0, nums))
print(result)
```

🎯 Tasks for Day 8

🧠 Task 1: Theory

1. What is function?
2. Difference between return and print
3. What is recursion?
4. What is lambda function?
5. What is map and filter?

💻 Task 2: Function Practice

1. Function to find factorial
2. Function to check prime number

3. Function to reverse string
4. Function to calculate average

💻 **Task 3: Recursion Practice**

1. Factorial using recursion
2. Sum of numbers using recursion
3. Fibonacci series

💻 **Task 4: Lambda & Map**

1. Square all numbers
2. Double all values
3. Convert list to uppercase

💻 **Task 5: Filter Practice**

1. Find even numbers
2. Filter students with marks > 50
3. Filter words with length > 5

🔥 **Task 6: Logic Challenge**

1. Find sum of digits using recursion
2. Flatten nested list
3. Count occurrences using dictionary